

CoRoT-1b

R-2612

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-1_150209 · created 2026-08-28T01:38:47+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457062.7035 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.027
Transit depth (Rp/Rs)^2	1.2e-05 +/- 0.0018 [%]
Orbital Inclination (inc)	86.12 +/- 2.86
Airmass coefficient 1 (a1)	0.954 +/- 0.02
Airmass coefficient 2 (a2)	0.035 +/- 0.035
Scatter in the residuals of the lightcurve fit is	2.08 %
Best Comparison Star	#1 - [107, 175]
Optimal Aperture	6.21
Optimal Annulus	15.86
Transit Duration (day)	0.094 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.0 ± 0.027 vs 0.1388 (deviation 5.14σ)
Duration fit vs published	0.094 d vs 0.1046 d (deviation 0.96σ)
Mid-time O-C	-7.6 min
Depth SNR	0.0
Residual scatter	2.08 %

Light curve

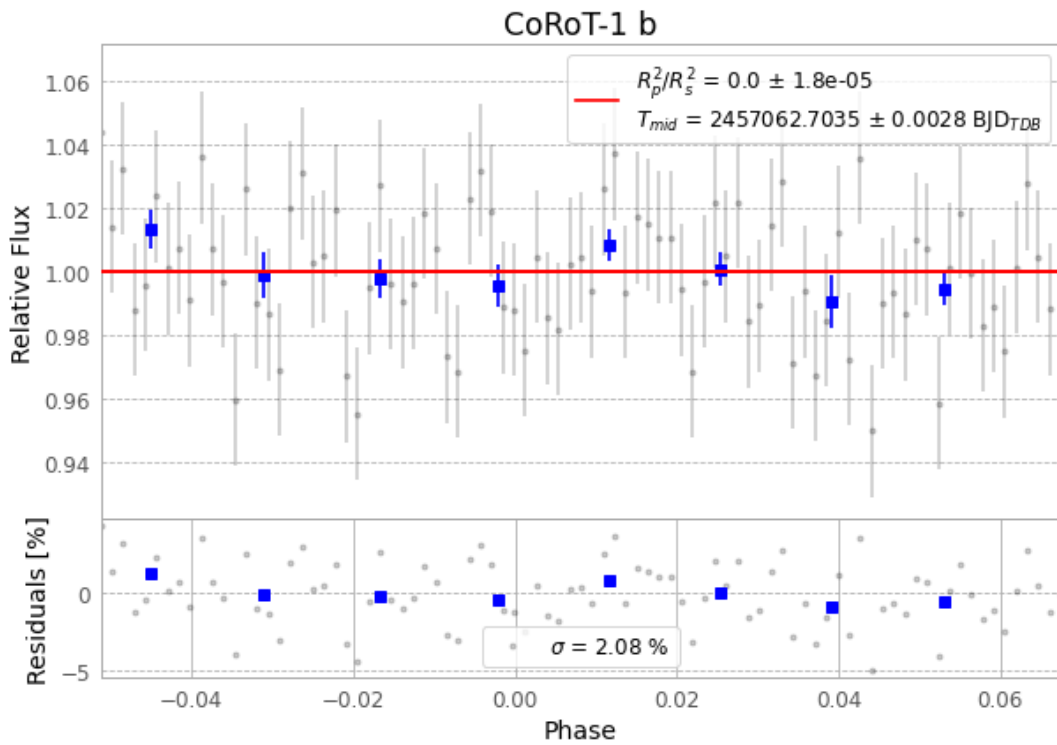


Figure 1 — FinalLightCurve_CoRoT-1 b_2015-02-08.png (SHA-256 20d8ea520e43cd9e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [90, 396], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fcd27e17c47d2850...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

87 FITS frames (57 MB), CoRoT-1150209025412.FITS → CoRoT-1150209071213.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-150209.log (2389f2860e31...), FinalLightCurve_CoRoT-1 b_2015-02-08.png (20d8ea520e43...), FinalLightCurve_CoRoT-1 b_2015-02-08.csv (36175fcede4d...)

Compute resources

Wall time 226.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-28 01:38Z.